

Ecommerce Product Data Analysis

Name : ABHAY CHARAN V
Intern ID : RL26030201
Course : Bachelor Of Technology
Internship Domain : Data Analytics
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##Introduction

In today's digital era, e-commerce platforms generate a vast amount of product data that can be analyzed to gain valuable insights. Understanding this data helps organizations make better decisions related to product management, pricing strategies, and inventory management. The main purpose of this project is to analyze an e-commerce product dataset and extract meaningful insights about product distribution, pricing patterns, and stock availability. By analyzing these factors, businesses can understand which products are more common, how prices are distributed, and how efficiently inventory is managed. In this project, Python is used along with data analysis libraries such as Pandas, Matplotlib, and Seaborn to process, analyze, and visualize the data. Various techniques are applied to explore the dataset and present the findings in a clear and understandable manner using graphs and charts.

##Objective of the project

The objective of this project is to analyze an e-commerce product dataset to identify patterns in product categories, brand distribution, pricing, and inventory levels. The goal is to derive meaningful insights that can help improve product management, pricing strategies, and stock planning.

##Tools Used

Python
Google Colab
Pandas
Matplotlib
Seaborn

##Dataset Description

The dataset contains 10,000 product records with 13 attributes, including product name, brand, category, price, stock levels, color, size, availability status, and unique identifiers such as EAN and Internal ID. This dataset is used to analyze product distribution, pricing patterns, and inventory management.

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px
```

```
df = pd.read_csv('products.csv')
```

```
df.head()
```

	Index	Name	Description	Brand	Category	Price	Currency	Stock	EAN	Color	Size	Availability
0	1	Smart Fan Iron Cooker Go Wireless Portable	Catch enough role nearly.	Herman Ltd	Kids' Clothing	585	USD	194	3968600833473	Cornsilk	5x7 in	limited_stock
1	2	Fan	All movement yeah tax me.	Braun, King and Rollins	Grooming Tools	992	USD	724	191126950284	Bisque	S	discontinued
2	3	Smart Speakerphone Charger Eco Plus Clean	Quickly inside pull line lay start.	Peck- Coleman	Fishing & Hunting	940	USD	769	7569143820621	Blue	Extra Large	pre_order
		Premium Grill	I awver one	Hines							50x70	

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
df.shape
```

```
(10000, 13)
```

```
df.columns
```

```
Index(['Index', 'Name', 'Description', 'Brand', 'Category', 'Price',
      'Currency', 'Stock', 'EAN', 'Color', 'Size', 'Availability',
      'Internal ID'],
      dtype='object')
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10000 entries, 0 to 9999
Data columns (total 13 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Index            10000 non-null  int64
1   Name             10000 non-null  object
2   Description       10000 non-null  object
3   Brand            10000 non-null  object
4   Category         10000 non-null  object
5   Price            10000 non-null  int64
6   Currency         10000 non-null  object
7   Stock            10000 non-null  int64
8   EAN              10000 non-null  int64
9   Color            10000 non-null  object
10  Size             10000 non-null  object
11  Availability      10000 non-null  object
12  Internal ID       10000 non-null  int64
dtypes: int64(5), object(8)
memory usage: 1015.8+ KB
```

```
df.isnull().sum()
```

	0
Index	0
Name	0
Description	0
Brand	0
Category	0
Price	0
Currency	0
Stock	0
EAN	0
Color	0
Size	0
Availability	0
Internal ID	0

dtype: int64

```
df.duplicated().sum()
```

```
np.int64(0)
```

```
df.describe()
```

	Index	Price	Stock	EAN	Internal ID
count	10000.00000	10000.000000	10000.000000	1.000000e+04	10000.000000
mean	5000.50000	503.370200	499.258000	5.035709e+12	50.475500
std	2886.89568	289.745777	287.993559	2.887653e+12	28.555993
min	1.00000	1.000000	1.000000	2.612759e+08	1.000000
25%	2500.75000	251.000000	249.000000	2.545172e+12	26.000000
50%	5000.50000	506.000000	500.000000	5.061908e+12	51.000000
75%	7500.25000	756.000000	751.000000	7.541982e+12	75.000000
max	10000.00000	999.000000	999.000000	9.999340e+12	99.000000

```
df['Price'].describe()
```

Price	
count	10000.000000
mean	503.370200
std	289.745777
min	1.000000
25%	251.000000
50%	506.000000
75%	756.000000
max	999.000000

dtype: float64

df['Brand'].value_counts().head()

count	
Brand	
Kennedy LLC	5
Buckley Inc	4
Watkins Group	4
Flores LLC	4
Johnston Ltd	4

dtype: int64

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px
```

df = pd.read_csv('products.csv')

df.head()

Index		Name	Description	Brand	Category	Price	Currency	Stock	EAN	Color	Size	Availability
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2	3	Smart Speakerphone Charger Eco Plus Clean	Quickly inside pull line lay start.	Peck-Coleman	Fishing & Hunting	940	USD	769	7569143820621	Blue	Extra Large	pre_order
		Premium Grill	Lawyer one	Hines								
												50x70

Next steps: [Generate code with df](#) [New interactive sheet](#)

df['Category'].value_counts()

	count
Category	
Clothing & Apparel	320
Kitchen Appliances	319
Home & Kitchen	316
Team Sports	316
Headphones & Earbuds	311
Grooming Tools	311
Smartwatches	310
Cameras & Accessories	310
Health & Wellness	310
Beauty & Personal Care	306
Camping & Hiking	306
Automotive	305
Home Decor	303
Books & Stationery	301
Fishing & Hunting	300
Shoes & Footwear	298
Men's Clothing	292
Haircare	292
Smartphones	292
Furniture	291
Skincare	290
Fragrances	287
Kids' Clothing	286
Makeup	286
Cleaning Supplies	284
Toys & Games	283
Cycling	279
Fitness Equipment	278
Women's Clothing	277
Office Supplies	275
Sports & Outdoors	272
Accessories (Bags, Hats, Belts)	266
Bedding & Bath	264
Laptops & Computers	264

dtype: int64

```
df['Brand'].value_counts().head(10)
```

count	
Brand	
Kennedy LLC	5
Buckley Inc	4
Watkins Group	4
Flores LLC	4
Johnston Ltd	4
Potter Group	4
Lewis LLC	4
Franco Inc	4
Fuentes LLC	4
Vaughn LLC	4

dtype: int64

df['Availability'].value_counts()

count	
Availability	
discontinued	1706
out_of_stock	1700
pre_order	1673
limited_stock	1644
in_stock	1644
backorder	1633

dtype: int64

low_stock = df[df['Stock'] <10]
low_stock.head()

Index		Name	Description	Brand	Category	Price	Currency	Stock	EAN	Color	Size	Availability
90	91	Compact Powerbank Radio Lite Digital	Professional kitchen move hold pay.	Brown-Berg	Beauty & Personal Care	455	USD	8	7242294927431	LightSeaGreen	Extra Large	in_
174	175	Advanced Dock Automatic Silent Portable	Hope star recently certain.	Brady PLC	Automotive	563	USD	1	1743619969434	LightBlue	Extra Large	out_of_
307	308	Smart Microphone Lamp Speaker Ultra	Value feel lawyer again.	Thomas LLC	Books & Stationery	757	USD	9	2578937487174	DarkSlateGray	XXL	limited_

Next steps: [Generate code with low_stock](#) [New interactive sheet](#)

out_of_stock = df[df['Stock'] ==0]
out_of_stock.head()

Index	Name	Description	Brand	Category	Price	Currency	Stock	EAN	Color	Size	Availability	Internal ID	
-------	------	-------------	-------	----------	-------	----------	-------	-----	-------	------	--------------	-------------	---

df['EAN'].duplicated().sum()

np.int64(0)

low_stock = df[df['Stock'] <10]
low_stock.shape

(84, 13)

```
out_of_stock = df[df['Stock'] ==0]
out_of_stock.shape
```

(0, 13)

```
df.to_csv('clean_products.csv', index=False)
```

```
from google.colab import files
files.download('clean_products.csv')
```

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df = pd.read_csv("clean_products.csv")
df.head()
```

	Index	Name	Description	Brand	Category	Price	Currency	Stock	EAN	Color	Size	Availability
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		Premium Grill	Lawver one	Hines							50x70	

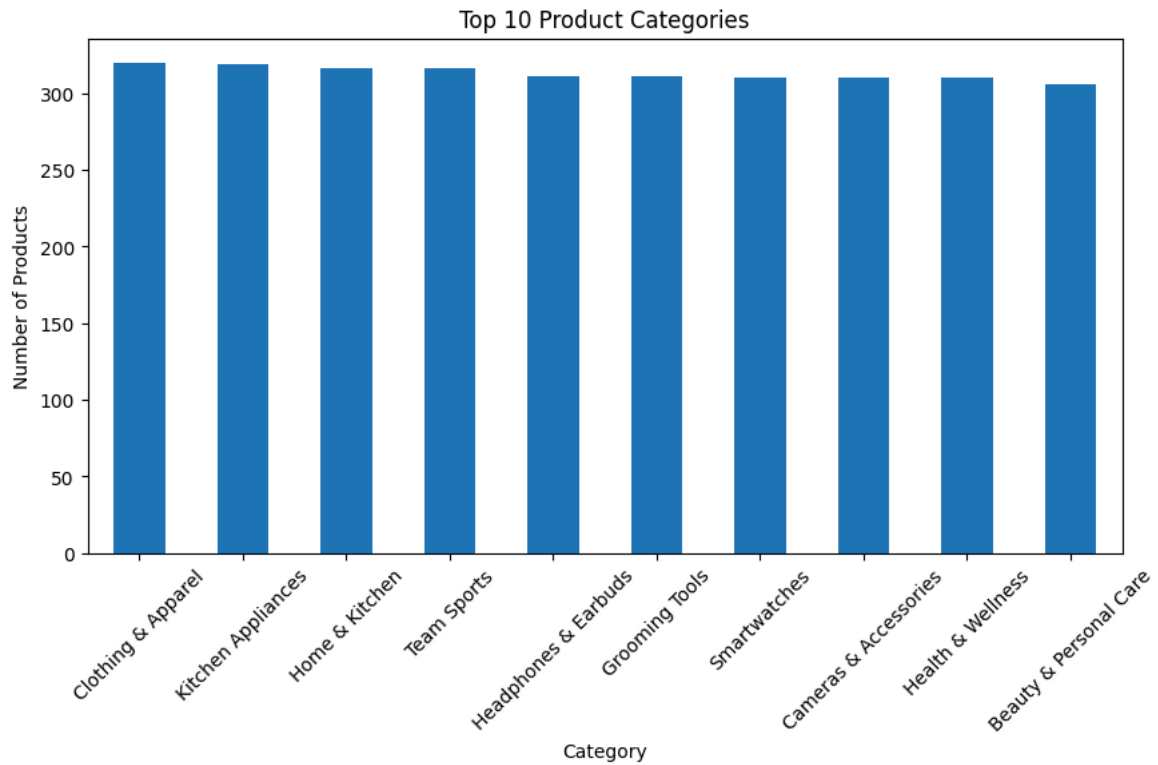
Next steps: [Generate code with df](#) [New interactive sheet](#)

```
df.info()
df.describe()
```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10000 entries, 0 to 9999
Data columns (total 13 columns):
Column Non-Null Count Dtype
--- -
0 Index 10000 non-null int64
1 Name 10000 non-null object
2 Description 10000 non-null object
3 Brand 10000 non-null object
4 Category 10000 non-null object
5 Price 10000 non-null int64
6 Currency 10000 non-null object
7 Stock 10000 non-null int64
8 EAN 10000 non-null int64
9 Color 10000 non-null object
10 Size 10000 non-null object
11 Availability 10000 non-null object
12 Internal ID 10000 non-null int64
dtypes: int64(5), object(8)
memory usage: 1015.8+ KB

	Index	Price	Stock	EAN	Internal ID
count	10000.00000	10000.000000	10000.000000	1.000000e+04	10000.000000
mean	5000.50000	503.370200	499.258000	5.035709e+12	50.475500
std	2886.89568	289.745777	287.993559	2.887653e+12	28.555993
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50%	5000.50000	506.000000	500.000000	5.061908e+12	51.000000
75%	7500.25000	756.000000	751.000000	7.541982e+12	75.000000
max	10000.00000	999.000000	999.000000	9.999340e+12	99.000000

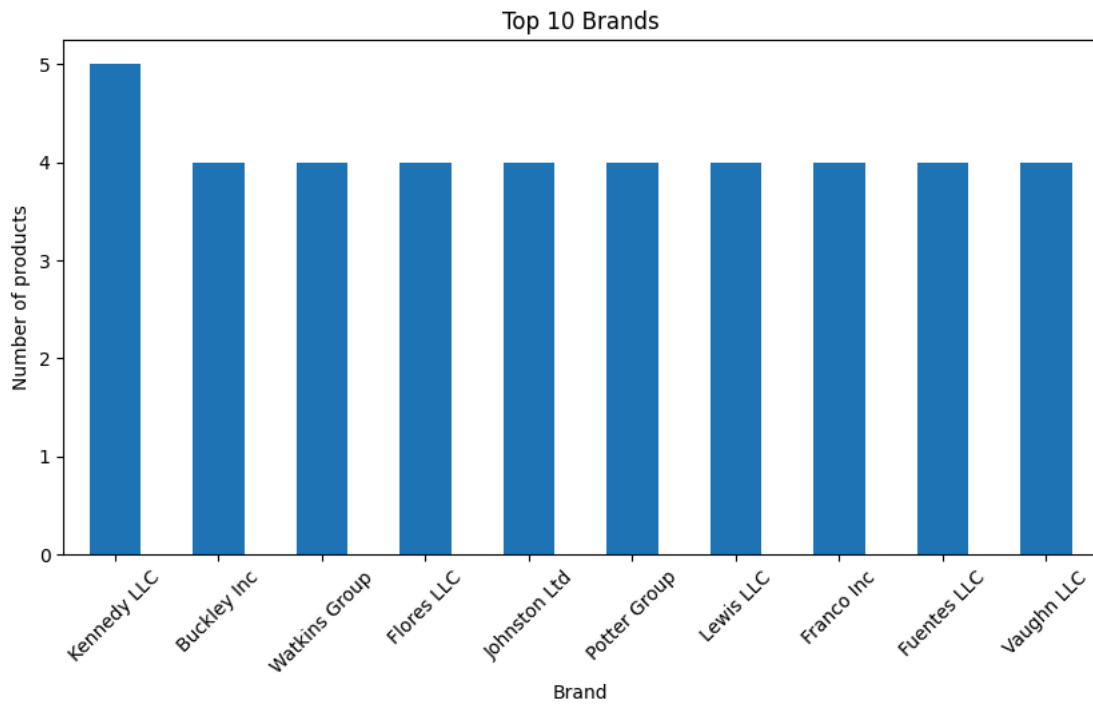
```
plt.figure(figsize=(10,5))
df['Category'].value_counts().head(10).plot(kind='bar')
plt.title('Top 10 Product Categories')
plt.xlabel('Category')
plt.ylabel('Number of Products')
plt.xticks(rotation=45)
plt.show()
```



Insight

The bar chart shows the Distribution of Products across different categories. Clothing and appeal has the hishest number of products, followed by Kitchen Appliances and Home & Kitchen. This indicates that these categories dominate the product inventory.

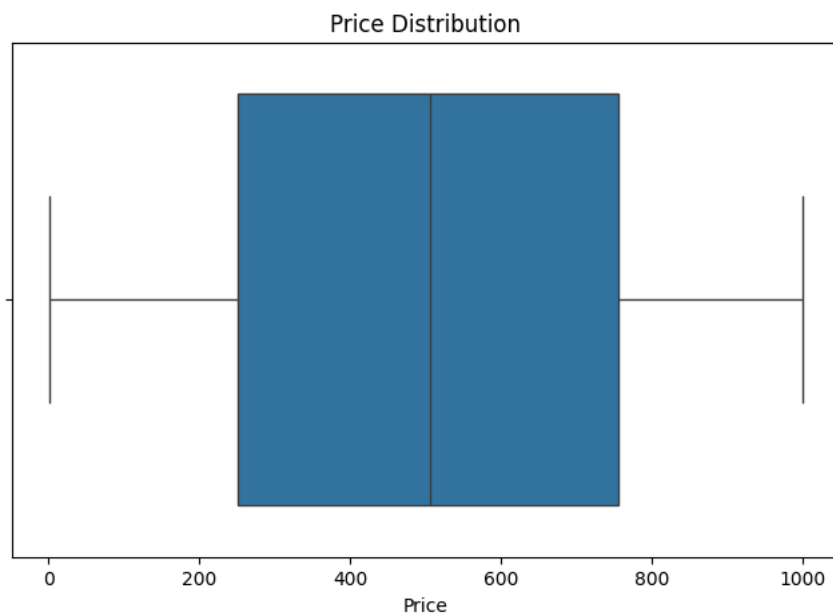
```
plt.figure(figsize=(10,5))
df['Brand'].value_counts().head(10).plot(kind='bar')
plt.title('Top 10 Brands')
plt.xlabel('Brand')
plt.ylabel('Number of products')
plt.xticks(rotation=45)
plt.show()
```



Insight

The analysis of brand distribution shows that no single brand dominates the dataset. The top brands have very similar product counts, indicating a balanced distribution of products across multiple brands.

```
plt.figure(figsize=(8,5))
sns.boxplot(x=df['Price'])
plt.title('Price Distribution')
plt.show()
```

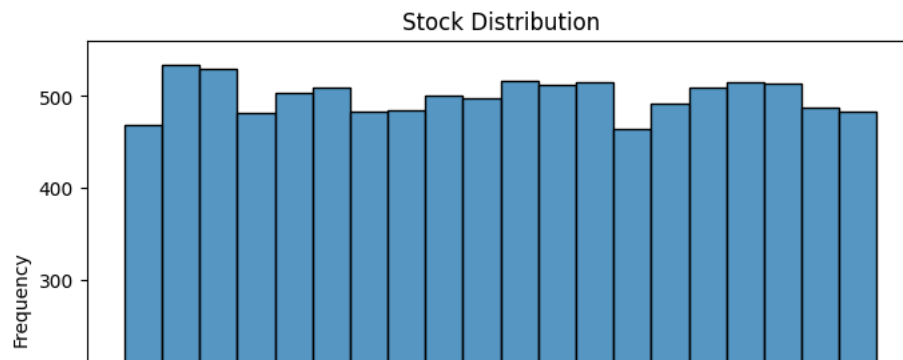


Insight

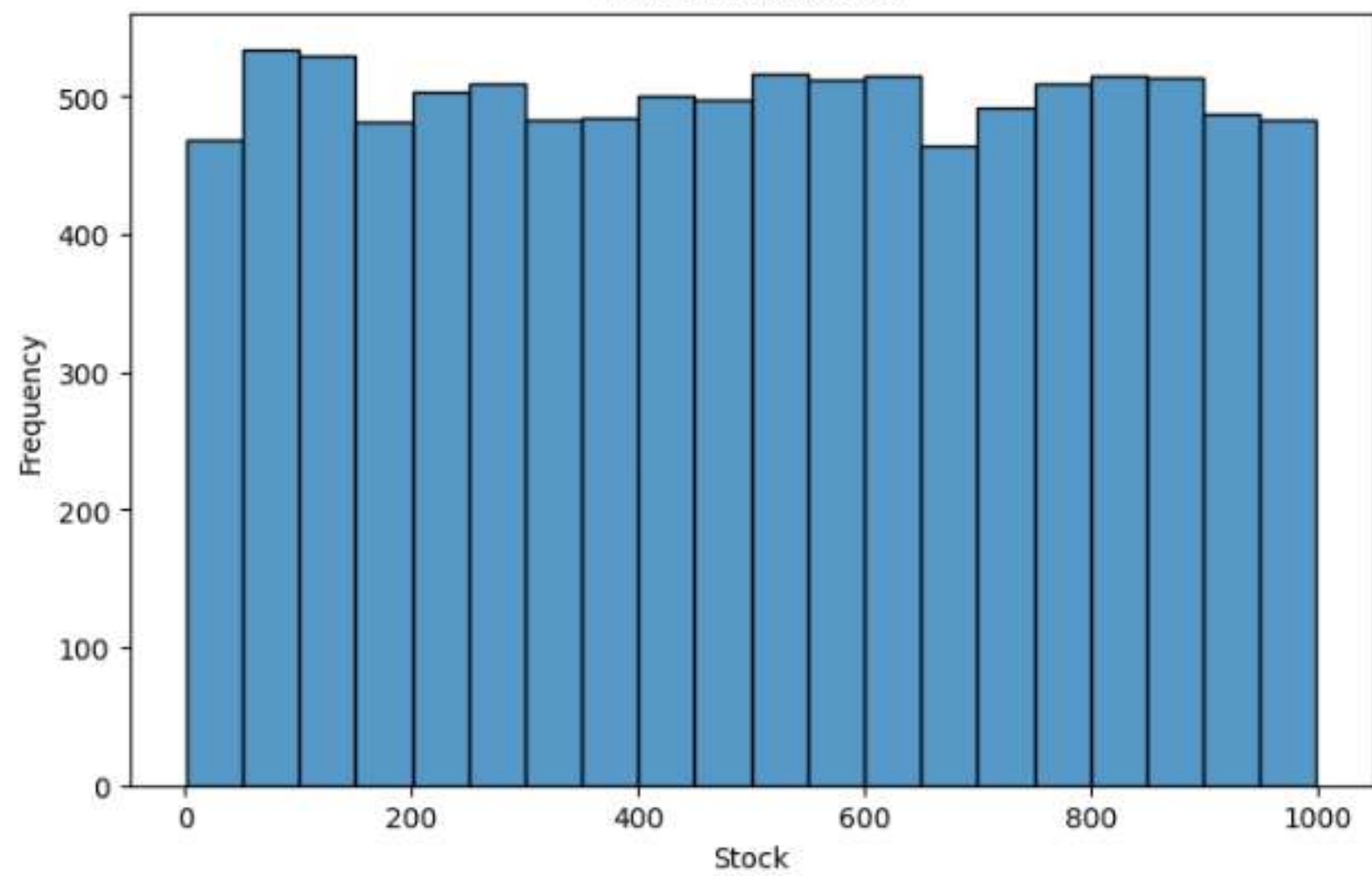
The boxplot shows that product prices are distributed between 1 and 999. The median price is around 500, indicating that most products are moderately priced. There are no extreme outliers, suggesting a uniform price distribution.

```
plt.figure(figsize=(8,5))
sns.histplot(df['Stock'], bins=20)
plt.title('Stock Distribution')
plt.xlabel('Stock')
```

```
plt.ylabel('Frequency')  
plt.show()
```



Stock Distribution



Insight

The stock distribution shows that most products have a moderate stock level. Very few products have extremely low stock, and no products are completely out of stock. This indicates good inventory management in the dataset.

Overall Analysis Insight

The analysis of the dataset shows that the product distribution is well balanced across categories and brands. Most products fall within a moderate price range, and inventory levels are generally well maintained. There are only a few low-stock products and no out-of stock items, indicating efficient stock management.

##Conclusion

In this project, we analyzed an e-commerce product dataset to understand product distribution, pricing trends, and inventory status. The analysis revealed that certain categories such as Clothing & Apparel dominate the dataset, while brand distribution is relatively balanced. Most products fall within a moderate price range, and inventory levels are well maintained, with very few low-stock items and no out-of-stock products. These insights can help businesses make better decisions related to product management, pricing strategies, and stock control.